

Two Mark Questions

UNIT I

Differential calculus

Problem 1. Define Radius of curvature in cartesian form and curvature?[May 2017, May 2016, Nov. 2016, Jan 2015]

The radius of curvature in Cartesian form is given by

$$\rho = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}{\frac{d^2y}{dx^2}} ..$$

$$\text{Curvature} = \frac{1}{\text{radius}}$$

Problem 2. Write the relation between beta and gamma function?[Jan 2015, Jan 2014, May 2014]

The relation between beta and gamma function is given by $\beta(m, n) =$

$$\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

Problem 3. Write down the formula for radius of curvature in terms of parametric co-ordinate system.[May 2014]

Solution. $\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \ddot{x}\dot{y}}$ where $\dot{x} = \frac{dx}{dt}, \dot{y} = \frac{dy}{dt}.$

Problem 4. For the curve $ay^2 = x^3$, find the curvature at (a, a)

Solution. $ay^2 \Rightarrow x^3 \Rightarrow 2a yy' = 3x^2$ and $2a[y'y' + yy''] = 6x$

Therefore, at (a, a) , $2a(a)y' = 3a^2$ and $2a[y'^2 + ay''] = 6a$ or $y'^2 + ay'' = 3$

$$\Rightarrow y' = \frac{3}{2} \text{ and } \left(\frac{3}{2}\right)^2 + ay'' = 3 \Rightarrow y'' = \frac{1}{a} \left[3 - \frac{9}{4}\right] = \frac{3}{4a}$$

$$\text{Hence } \rho = \frac{(1+y'^2)^{3/2}}{y''} = \frac{\left(1+\frac{9}{4}\right)^{3/2}}{3/4a} = \frac{4a}{3} \left(\frac{13}{4}\right)^{3/2} = \frac{4a}{3} \left(\frac{13\sqrt{13}}{4\sqrt{4}}\right)$$

Therefore, the curvature at (a, a) is $\frac{1}{\rho} = \frac{6}{13\sqrt{13}a}$

Problem 5. What is the radius of curvature of the curve $x^4 + y^4 = 2$

Solution. Differentiating $x^4 + y^4 = 2$ w.r.t $x \Rightarrow 4x^3 + 4y^3 y' = 0$
 $2 \Rightarrow y' = -\frac{x^3}{y^3}$ (1)

Differentiating $x^3 + y^3 y' = 0$ we get $3x^2 + 3y^2 y'^2 + y^3 y'' = 0 \Rightarrow$
 $y^3 y'' = -3x^2 - 3y^2 (y')^2$
 $\Rightarrow y'' = \frac{-3x^2 - 3y^2 (y')^2}{y^3}$ (2)

At $(1, 1)$, from (1), $y' = -1$ and from (2), $y'' = \frac{-3 - 3(-1)^2}{1} = -6$

Therefore $\rho = \frac{(1 + y'^2)^{3/2}}{y''} = \frac{(1 + 1)^{3/2}}{-6} = \frac{-2\sqrt{2}}{6} = \frac{-\sqrt{2}}{3}$

$\Rightarrow \frac{\sqrt{2}}{3}$ (numerically).

Problem 6. Find the radius of curvature for $y = e^x$ at the point where it cuts the y-axis.

Solution. At the point where $y = e^x$ cuts the y-axis, $x = 0$
 $x = 0 \Rightarrow y = e^0 = 1$. Therefore $y = e^x$ cuts the y-axis at $(0, 1)$

$$y = e^x \Rightarrow y' = e^x \text{ and } y'' = e^x$$

Therefore, at $(0, 1)$, $y' = e^0 = 1$ and $y'' = e^0 = 1$

$$\text{Hence, } \rho = \frac{(1 + y'^2)^{3/2}}{y''} = \frac{(1+1)^{3/2}}{1} = 2^{3/2} = 2\sqrt{2}.$$

Problem 7. Find the radius of the curve $xy = c^2$ at (c, c) .

$$\textbf{Solution. } xy = c^2 \Rightarrow y = \frac{c^2}{x} \Rightarrow y' = \frac{-c^2}{x^2} \text{ and } y'' = \frac{2c^2}{x^3}$$

$$\text{At } (c, c), y' = \frac{-c^2}{c^2} = -1 \text{ and } y'' = \frac{2c^2}{c^3} = \frac{2}{c}$$

$$\text{Therefore } \rho = \frac{(1 + y'^2)^{3/2}}{y''} = \frac{(1 + 1)^{\frac{3}{2}}}{\frac{2}{c}} = \frac{c(2)^{\frac{3}{2}}}{2} = \sqrt{2}c.$$

Problem 8. Find the radius of the curve $x = \frac{\pi}{2}$ on the curve $y = 4 \sin x$.

$$\textbf{Solution. } y = 4 \sin x \Rightarrow y' = 4 \cos x, y'' = -4 \sin x$$

$$\text{At } x = \frac{\pi}{2}, y' = 4 \cos \frac{\pi}{2} = 0 \text{ and } y'' = -4 \sin \frac{\pi}{2} = -4$$

$$\text{Therefore, } \rho = \frac{(1+y'^2)^{3/2}}{y''} = \frac{-1}{4} \text{ i.e., } \rho = \frac{1}{4} \text{ (numerically).}$$

Problem 9. Find the radius of the curvature of $x = \frac{\pi}{2}$ on the curve $y = 4 \sin x - \sin 2x$.

$$\textbf{Solution. } y = 4 \sin x - \sin 2x, \Rightarrow y' = 4 \cos x - 2 \cos 2x, \Rightarrow y'' = -4 \sin x + 4 \sin 2x$$

$$\text{Therefore at } x = \frac{\pi}{2}, y' = 4 \cos \frac{\pi}{2} - 2 \cos \pi = 4(0) - 2(-1) = 2$$

$$y'' = -4 \sin \frac{\pi}{2} + 4 \sin \pi = -4(1) + 4(0) = -1$$

$$\text{Hence, } \rho = \frac{(1+y'^2)^{3/2}}{y''} = \frac{(1+4)^{3/2}}{-1} = \frac{5^{3/2}}{-1} = \frac{-5\sqrt{5}}{1} \text{ i.e., } \rho = \frac{5\sqrt{5}}{1} \text{ (numerically).}$$

Problem 10. Find the radius of the curvature of $y = \log(\sec x)$

at any point on it.

Solution. $y = \log(\sec x) \Rightarrow y' = \frac{1}{\sec x} \sec x \tan x = \tan x$ and $y'' = \sec^2 x$

Therefore, $\rho = \frac{(1 + y'^2)^{3/2}}{y''} = \frac{(1 + \tan^2 x)^{3/2}}{-4} = \frac{(\sec^2 x)^{3/2}}{\sec^2 x} = \frac{\sec^3 x}{\sec^2 x} = \sec x$.

Problem 11. Find the radius of the curvature of $y = x \cosh\left(\frac{x}{c}\right)$ at $(0, c)$.

Solution. $y = c \cosh\left(\frac{x}{c}\right) \Rightarrow y' = c \sinh\left(\frac{x}{c}\right) \frac{1}{c} = \sinh\left(\frac{x}{c}\right)$ and $y'' = \frac{1}{c} \cosh\left(\frac{x}{c}\right)$

Therefore at $(0, c)$ $y' = \sinh(0)$ and $y'' = \frac{1}{c} \cosh(0) = \frac{1}{c}(1) = \frac{1}{c}$.

$$\rho = \frac{(1 + y'^2)^{3/2}}{y''} = \frac{(1 + 0^2)^{3/2}}{1/c} = c$$

Problem 12. Find the curvature $(3, -4)$ to the curve $x^2 + y^2 = 25$

Solution. $x^2 + y^2 = 25$ is a circle of radius 5

Solution. Therefore, the radius of curvature is 5 and hence the curvature at $(3, -4)$ is $\frac{1}{5}$.

Problem 13. Find the radius of curvature of the curve $x = t^2, y = t$ at $t = 1$

Solution. $\dot{x} = 2t, \dot{y} = 1, \ddot{x} = 2, \ddot{y} = 0 \Rightarrow$ At $t = 1, \dot{x} = 2, \dot{y} = 1, \ddot{x} = 2$.

Hence $\rho = \frac{(\dot{x}^2 + \dot{y}^2)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}} = \frac{(4 + 1)^{3/2}}{2(0) - (1)(2)} = \frac{(5)^{3/2}}{-2} \Rightarrow \rho = \frac{5\sqrt{5}}{2}$ numerically.

Problem 14. If the center of curvature of a curve at a variable point 't' on it is $\left(\frac{c}{a} \cos^3 t, \frac{-c}{b} \sin^3 t\right)$, find the evolute of the curve.

Solution. $\bar{x} = \frac{c}{a} \cos^3 t$ and $\bar{y} = \frac{-c}{b} \sin^3 t$

$$\Rightarrow \cos^3 t = \frac{a\bar{x}}{c} \text{ and } \sin^3 t = \frac{-b\bar{y}}{c}$$

$$\Rightarrow (\cos^3 t)^{2/3} = \left(\frac{a\bar{x}}{c}\right)^{2/3} \text{ and } (\sin^3 t)^{2/3} = \left(\frac{-b\bar{y}}{c}\right)^{2/3}$$

$$\Rightarrow \cos^2 t = \left(\frac{a\bar{x}}{c}\right)^{2/3} \text{ and } \sin^2 t = \left(\frac{b\bar{y}}{c}\right)^{2/3}$$

$$\text{Therefore, } \cos^2 t + \sin^2 t = 1 \Rightarrow \left(\frac{a\bar{x}}{c}\right)^{2/3} + \left(\frac{b\bar{y}}{c}\right)^{2/3} = 1$$

$$\Rightarrow (a\bar{x})^{2/3} + (b\bar{y})^{2/3} = c^{2/3}$$

Hence the evolute is $(ax)^{2/3} + (by)^{2/3} = c^{2/3}$.

Problem 15. Define the circle of curvature at a point $P(x_1, y_1)$ on the curve $y = f(x)$

Solution. Equation of the circle of curvature at (X_1, y_1) is $(x - \bar{x})^2 + (y - \bar{y})^2 = \rho^2$, where $(\bar{x}, \bar{y})$ is center of curvature and ρ is the radius of curvature of the curve $y = f(x)$ at $P(x_1, y_1)$

Problem 16. Find the curvature of the curve given by $S = c \tan \psi$ at $\psi = 0$

Solution. $S = c \tan \psi \Rightarrow \frac{ds}{d\psi} = c \sec^2 \psi$. At $\psi = 0$, $\frac{ds}{d\psi} = c \sec^2 0 = c(1)^2 = c$

Therefore, the curvature of the curve at $\psi = 0$ is $\frac{ds}{d\psi} = \frac{1}{c}$

Problem 17. Define Radius of curvature in parametric form?

The radius of curvature in parametric form is given by

$$\rho = \frac{[(x')^2 + (y')^2]^{\frac{3}{2}}}{(x')(y'') - (y')(x'')}.$$

where $x' = \frac{dx}{dt}$, $x'' = \frac{d^2x}{dt^2}$, $y' = \frac{dy}{dt}$ and $y'' = \frac{d^2y}{dt^2}$.

Problem 18. Define Radius of curvature in Polar form?

The radius of curvature in polar form is given by $\rho = \frac{(r^2 + r'^2)^{\frac{3}{2}}}{r^2 - rr'' + 2r'^2}$.

Problem 19. Define center of curvature?

The co-ordinates of the center of curvature is given by $\bar{X} =$

$$x - \frac{\frac{dy}{dx} \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}}{\frac{d^2y}{dx^2}} \text{ and } \bar{Y} = y + \left\{ \frac{1 + \left(\frac{dy}{dx} \right)^2}{\frac{d^2y}{dx^2}} \right\}$$

Problem 20. Define Evolute?[Jan 2016]

The locus of centre of curvature of any curve is called its evolute.

Problem 21. Define Envelope of a family of curve?

A curve which touches the each member of a family of curve is called Envelope.

Problem 22. Define Involute?[Jan 2016]

The involute of a given curve is the locus of the free end of a line segment whose another end is tangent to the given curve. This line segment is wound or reversely unwound along with the

continuous variation in the length of line segment by a certain amount. The curve so traced by free end is called an involutes.

Problem 23. Define Gamma Function?

The definite integral $\int_0^\infty e^{-x} x^{n-1} dx$ exists only when $n > 0$ and when it exists, it is a function of n and called Gamma function and denoted by $\Gamma(n)$. Thus $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$.

Problem 24. Define beta Function?

The definite integral $\int_0^\infty x^{m-1} (1-x)^{n-1} dx$ exists only when $m > 0$ and when it exists, it is a function of m and n and called beta function and denoted by $\beta(m, n)$. Thus $\beta(m, n) = \int_0^\infty x^{m-1} (1-x)^{n-1} dx$.

Problem 25. Find the curvature of the curve $2x^2 + 2y^2 + 5x - 2y + 1 = 0$.

Solution. Given curve is the circle $x^2 + y^2 + \frac{5}{2}x - y + \frac{1}{2} = 0$.

Hence the Radius = $\sqrt{g^2 + f^2 - c} = \sqrt{\left(\frac{5}{4}\right)^2 + \left(\frac{-1}{2}\right)^2 - \frac{1}{2}} =$
 $\sqrt{\frac{25 + 4 - 8}{16}} = \sqrt{\frac{21}{16}} = \frac{\sqrt{21}}{4}$. Curvature = $\frac{1}{radius} = \frac{4}{\sqrt{21}}$

UNIT II

Functions of Several variables

Problem 1. Find $\frac{du}{dx}$, if $u = \tan^{-1} \left(\frac{y}{x} \right)$ where $y = \tan^2 x$

Solution.

$$\begin{aligned} du &= \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \Rightarrow \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx} \\ &= \frac{1}{1 + \frac{y^2}{x^2}} \left(\frac{-y}{x^2} \right) + \frac{1}{1 + \frac{y^2}{x^2}} \left(\frac{1}{x} \right) \cdot 2 \tan x \sec^2 x \\ &= \frac{-y}{x^2 + y^2} + \frac{2x \sec^2 x \tan x}{x^2 + y^2} = \frac{2x \sec^2 x \tan x - y}{x^2 + y^2} \end{aligned}$$

Problem 2. If $u = \frac{y^2}{x}$, $v = \frac{x^2}{y}$ find $\frac{\partial(x, y)}{\partial(u, v)}$

Solution.

$$\begin{aligned} \frac{\partial(x, y)}{\partial(u, v)} dx &= \frac{1}{\left(\frac{\partial(u, v)}{\partial(x, y)} \right)} \\ \frac{\partial u}{\partial x} &= \frac{-y^2}{x^2}, \frac{\partial u}{\partial y} = \frac{2y}{x}, \frac{\partial v}{\partial x} = \frac{2x}{y}, \frac{\partial v}{\partial y} = \frac{-x^2}{y^2} \\ \frac{\partial(u, v)}{\partial(x, y)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{-y^2}{x^2} & \frac{2y}{x} \\ \frac{2x}{y} & \frac{-x^2}{y^2} \end{vmatrix} = 1 - 4 = -30 \end{aligned}$$

Problem 3. Using Euler's theorem, given $u(x, y)$ is a homogeneous function of degree n , prove that $x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n(n-1)u$

Solution.

By Euler's theorem

$$xu_x + yu_y = nu \quad (1)$$

Diff. (1) Partially w.r.t x

$$u_x + xu_{xx} + yu_{xy} = nu_x \quad (2)$$

Diff. (1) Partially w.r.t y

$$xu_{xy} + u_y + yu_{yy} = nu_y \quad (3)$$

$x \times (2) + y \times (3)$ gives

$$\begin{aligned} xu_x + yu_y + x^2u_{xx} + xyu_{xy} + y^2u_{yy} &= n(xu_x + yu_y) \\ \Rightarrow nu + [x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}] &= n(nu) \\ \Rightarrow [x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}] &= n^2u - nu \\ &= n(n-1)u \end{aligned}$$

Problem 4. Given $u(x, y) = x^2 \tan^{-1} \left(\frac{y}{x} \right)$, find the value of $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$

Solution.

u is a homogeneous function of degree 2 in x, y

$$\begin{aligned} \Rightarrow x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} &= n(n-1)u \\ &= 2(2-1)u \\ &= 2u \end{aligned}$$

Problem 5. Find $\frac{du}{dt}$ if $u = \sin \left(\frac{x}{y} \right)$ where $x = e^t, y = t^2$

Solution.

$$\begin{aligned}
\frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = \frac{1}{y} \cos\left(\frac{x}{y}\right) \cdot e^t + \left(\frac{-x}{y^2}\right) \cos\left(\frac{x}{y}\right) (2t) \\
&= \frac{1}{t^2} \cos\left(\frac{e^t}{t^2}\right) e^t - \frac{e^t}{t^4} \cos\left(\frac{e^t}{t^2}\right) (2t) \\
&= \frac{e^t}{t^2} \cos\left(\frac{e^t}{t^2}\right) - \frac{2e^t}{t^3} \cos\left(\frac{e^t}{t^2}\right) = \frac{\cos\left(\frac{e^t}{t^2}\right)}{t^2} \left(1 - \frac{2}{t}\right) e^t
\end{aligned}$$

Problem 6. If $u = \frac{y^2}{2x}$, $v = \frac{x^2 + y^2}{2x}$ find $\frac{\partial(u, v)}{\partial(x, y)}$

Solution.

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{-y^2}{2x^2} \\
\frac{\partial u}{\partial y} &= \frac{2y}{2x} = \frac{y}{x} \\
\frac{\partial v}{\partial x} &= \frac{2x(2x) - (x^2 + y^2)(2)}{4x^2} = \frac{2(x^2 - y^2)}{4x^2} = \frac{(x^2 - y^2)}{2x^2} \\
\frac{\partial v}{\partial y} &= \frac{2y}{2x} = \frac{y}{x}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial(u, v)}{\partial(x, y)} &= \begin{vmatrix} \frac{-y^2}{2x^2} & \frac{y}{x} \\ \frac{x^2 - y^2}{2x^2} & \frac{y}{x} \end{vmatrix} = \frac{-y^3}{2x^3} - \frac{y}{x} \left(\frac{x^2}{2x^2} - \frac{y^2}{2x^2} \right) \\
&= \frac{-y^3}{2x^3} - \frac{xy}{2x^2} + \frac{y^3}{2x^3} = -\frac{y}{2x}
\end{aligned}$$

Problem 7. If $u = \frac{x^3 y^2}{2x} + x^2 y^3$ where $x = at^2$ and $y = 2at$ then find $\frac{du}{dt}$

Solution.

$$\begin{aligned}
 \frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} = (3x^2y^2 + 2xy^3)(2at) + (2x^3y + 3x^2y^2)(2a) \\
 &= xy^2(3x + 2y)(2at) + x^2y(2x + 3y)(2a) \\
 &= 4a^3t^4(3at^2 + 4at)(2at) + (2a^3t^5)(2at^2 + 6at)(2a) \\
 &= 8a^4t^5(3at^2 + 4at) + 4a^4t^5(2at^2 + 6at) \\
 &= 24a^5t^7 + 32a^5t^6 + 8a^5t^7 + 24a^5t^6 \\
 &= 32a^5t^7 + 56a^5t^6 = 8a^5t^6(4t + 7)
 \end{aligned}$$

Problem 8. If $u = 2xy, v = x^2 - y^2, x = r \cos \theta, y = r \sin \theta$ then compute $\frac{\partial(u, v)}{\partial(r, \theta)}$ [May 2017]

Solution.

$$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)} \quad (1)$$

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2y & 2x \\ 2x & -2y \end{vmatrix} = -4(x^2 + y^2) = -4r^2$$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$$

From (1) $\frac{\partial(u, v)}{\partial(r, \theta)} = -4r^3$

Problem 9. Using the definition of total derivative, find the value of $\frac{du}{dt}$ given $u = y^2 - 4ax; x = at^2, y = 2at$

Solution.

$$\begin{aligned}\frac{du}{dt} &= \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt} \\ &= -4a(2at) + 2y(2a) = -8a^2t + 2(2at)(2a) \\ &= -8a^2t + 8a^2t = 0\end{aligned}$$

Problem 10. If $u = f(x - y, y - z, z - x)$, find $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$

Solution.

$$u = f(p, q, r) \text{ where } p = x - y, q = y - z, r = z - x.$$

Therefore

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} \quad (1)$$

$$\Rightarrow \frac{\partial u}{\partial x} = \frac{\partial f}{\partial p}(1) + \frac{\partial f}{\partial q}(0) + \frac{\partial f}{\partial r}(-1) = \frac{\partial f}{\partial p} - \frac{\partial f}{\partial r} \quad (2)$$

Similarly,

$$\frac{\partial u}{\partial y} = \frac{\partial f}{\partial q} - \frac{\partial f}{\partial p}, \frac{\partial u}{\partial z} = \frac{\partial f}{\partial r} - \frac{\partial f}{\partial q} \quad (3)$$

$$(1) + (2) + (3) \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

Problem 11. Find Taylor's series expansion of $e^x \sin y$ near the point $\left(-1, \frac{\pi}{4}\right)$ upto first degree terms.

Solution.

Let $f(x, y) = e^x \sin y$. Then $f_x = e^x \sin y$ and $f_y = e^x \cos y$

$$f\left(-1, \frac{\pi}{4}\right) = \frac{e^{-1}}{\sqrt{2}},$$

$$f_x\left(-1, \frac{\pi}{4}\right) = e^{-1} \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}e^{-1}$$

$$f_y\left(-1, \frac{\pi}{4}\right) = e^{-1} \cos\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}e^{-1}$$

Hence,

$$f(x, y) = f\left(-1, \frac{\pi}{4}\right) + \frac{1}{1!} \left[(x+1)f_x\left(-1, \frac{\pi}{4}\right) + \left(y - \frac{\pi}{4}\right)f_y\left(-1, \frac{\pi}{4}\right) \right] + \dots$$

$$\Rightarrow e^x \sin y = \frac{e^{-1}}{\sqrt{2}} + \left[(x+1)\frac{e^{-1}}{\sqrt{2}} + \left(y - \frac{\pi}{4}\right)\frac{e^{-1}}{\sqrt{2}} \right] + \dots$$

Problem 12. Find $\frac{du}{dx}$, when $u = \sin(x^2 + y^2)$ and $x^2 + 4y^2 = 9$

Solution.

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy \Rightarrow \frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx}$$

$$\Rightarrow \frac{du}{dx} = 2x \cos(x^2 + y^2) + 2y \cos(x^2 + y^2) \left[\frac{-2x}{8y} \right]$$

$$= 2x \cos(x^2 + y^2) - \frac{x}{2} \cos(x^2 + y^2)$$

$$= \frac{3x}{2} \cos(x^2 + y^2)$$

Problem 13. If $x = u(1 - v)$ and $y = 2uv$, find $\frac{\partial(x, y)}{\partial(u, v)}$

Solution.

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1-v & -u \\ 2v & 2u \end{vmatrix} = 2u - 2uv + 2uv = 2u$$

Problem 14. If $u = xe^yz$ where $y = 2x$ and $z = \sin x$, find $\frac{du}{dx}$

Solution.

$$\begin{aligned} \frac{du}{dx} &= \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx} + \frac{\partial u}{\partial z} \frac{dz}{dx} \\ \Rightarrow \frac{du}{dx} &= e^yz + xe^yz \cdot 2 + xe^y \cos x \\ &= e^y(z + 2xz + x \cos x) \\ &= e^2x(z + 2x \sin x + x \cos x) \end{aligned}$$

Problem 15. If $x = r \cos \theta$, $y = r \sin \theta$ find $\frac{\partial(x, y)}{\partial(r, \theta)}$ [Jan 2016, Jan 2015, Nov. 2016]

Solution.

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$$

Problem 16. If u, v, w are functions of independent variables x, y, z and $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$, find the value of $\frac{\partial(2u, 2v, 2w)}{\partial(x, y, z)}$

Solution.

$$\frac{\partial(2u, 2v, 2w)}{\partial(x, y, z)} = \begin{vmatrix} 2\frac{\partial u}{\partial x} & 2\frac{\partial u}{\partial y} & 2\frac{\partial u}{\partial z} \\ 2\frac{\partial v}{\partial x} & 2\frac{\partial v}{\partial y} & 2\frac{\partial v}{\partial z} \\ 2\frac{\partial w}{\partial x} & 2\frac{\partial w}{\partial y} & 2\frac{\partial w}{\partial z} \end{vmatrix} = 8 \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = 8(4) = 32$$

Problem 17. If $u = x + y$ and $y = uv$, find the Jacobian $\frac{\partial(x, y)}{\partial(u, v)}$

Solution. [Similar in May 2016]

$$u = x + y \Rightarrow u = x + uv \Rightarrow x = u - uv$$

Therefore

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 - v & -u \\ v & u \end{vmatrix} = u - uv + uv = u$$

Problem 18. Find the stationary points of $x^2 - xy + y^2 - 2x + y$ [Jan 2016]

Solution.

Let $f = x^2 - xy + y^2 - 2x + y$. Then $f_x = 2x - y - 2$ and $f_y = -x + 2y + 1$

Stationary points are given by $f_x = 0$ and $f_y = 0$. Therefore,

$$2x - y = 2 \quad (1)$$

$$-x + 2y = -1 \quad (2)$$

$$2 \times (2) \rightarrow 2x + 4y = -2 \quad (3)$$

$$(1) + (3) \rightarrow 3y = 0 \Rightarrow y = 0 \Rightarrow x = 1$$

$(1, 0)$ is a stationary point.

Problem 19. Find the stationary points of the function $f(x, y) = x^3 + y^2 - 12xy$

Solution.

$$f_x = 3x^2 - 12y \text{ and } f_y = 2y - 12x$$

Stationary points are given by $f_x = 0$ and $f_y = 0 \Rightarrow 3x^2 - 12y = 0$ and $2y - 12x = 0$

$$\Rightarrow x^2 = 4y \text{ and } y = 6x$$

$$\Rightarrow x^2 = 24x \Rightarrow x^2 - 24x = 0 \Rightarrow x(x - 24) = 0 \Rightarrow x = 0 \text{ (or) } x = 24$$

$$x = 0 \Rightarrow y = 0 \text{ and } x = 24 \Rightarrow y = 144$$

Therefore stationary points are $(0, 0)$ and $(24, 144)$.

Problem 20. Write the sufficient condition for $f(x, y)$ to have a maximum value at (a, b)

Solution.

$$f_x = 3x^2 - 12y \text{ and } f_y = 2y - 12x$$

At (a, b) , $f_x = f_y = 0$, $A = f_{xx} < 0$ and $\Delta = AC - B^2 > 0$ where $B = f_{xy}$ and $C = f_{yy}$

Problem 21. Write down the Taylor's series expansion of $f(x + h, y + k)$ in a series of (i) of h and k (ii) powers of x and y

Solution.

$$\begin{aligned}
 (i) \quad f(x+h, y+k) &= f(x, y) + \frac{1}{1!} \left[h \frac{\partial f(x, y)}{\partial x} + k \frac{\partial f(x, y)}{\partial y} \right] \\
 &+ \frac{1}{2!} \left[h^2 \frac{\partial^2 f(x, y)}{\partial x^2} + 2hk \frac{\partial^2 f(x, y)}{\partial x \partial y} \right. \\
 &\left. + k^2 \frac{\partial^2 f(x, y)}{\partial y^2} \right] + \dots
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad f(x+h, y+k) &= f(h, k) + \frac{1}{1!} \left[x \frac{\partial f(h, k)}{\partial x} + y \frac{\partial f(h, k)}{\partial y} \right] \\
 &+ \frac{1}{2!} \left[x^2 \frac{\partial^2 f(h, k)}{\partial x^2} + 2xy \frac{\partial^2 f(h, k)}{\partial x \partial y} \right. \\
 &\left. + y^2 \frac{\partial^2 f(h, k)}{\partial y^2} \right] + \dots
 \end{aligned}$$

Problem 22. Write down the Taylor's series expansion of $f(x, y)$ near the point (a, b)

Solution.

$$\begin{aligned}
 f(x, y) &= f(a, b) + \frac{1}{1!} \left[(x-a) \frac{\partial f}{\partial x}(a, b) + (y-b) \frac{\partial f}{\partial y}(a, b) \right] \\
 &+ \frac{1}{2!} \left[(x-a)^2 \frac{\partial^2 f}{\partial x^2}(a, b) + (y-b) \frac{\partial^2 f}{\partial x \partial y}(a, b) \right. \\
 &\left. + (y-b)^2 \frac{\partial^2 f}{\partial y^2}(a, b) \right]
 \end{aligned}$$

Problem 23. State any three properties of Jacobians[May 2014]

Solution.

(i) If u and v are functions of x and y ,

$$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(u, v)} = 1$$

(ii) If u and v are functions of x, y and if x, y are functions of r, θ ,

$$\frac{\partial(u, v)}{\partial(r, \theta)} = \frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(r, \theta)}$$

(iii) If u, v, w are functionally dependent functions of three independent variables x, y, z then

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$$

Note (1) The functions u, v, w are said to be functionally dependent if $F(u, v, w) = 0$. (2) Converse of Property. (3) is also true.

Problem 24. Define the relative maximum and relative minimum of a function of two variables

Solution.

A function $f(x, y)$ is said to have relative max. at (a, b) if $f(a, b) > f(a + h, b + k)$ for all small values of h and k

A function $f(x, y)$ is said to have relative min. at (a, b) if $f(a, b) < f(a + h, b + k)$ for all small values of h and k

Problem 25. State the conditions for the stationary point (a, b) of $f(x, y)$ to be (i) a maximum point (ii) a minimum point (iii) a saddle point

Solution.

Let $A = f_{xx}(a, b)$, $B = f_{xy}(a, b)$, $C = f_{yy}$ and $\Delta = AC - B^2$. Then

- (i) (a, b) is a max. point if $\Delta > 0$ and $A < 0$
- (ii) (a, b) is a min. point if $\Delta > 0$ and $A > 0$
- (iii) (a, b) is a saddle point if $\Delta < 0$

UNIT III

Multiple Integrals

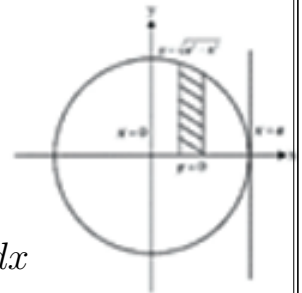
Problem 1. Evaluate $\int \int xy dx dy$ taken over the positive quadrant of the circle $x^2 + y^2 = a^2$.

Solution.

$$\iint_A xy dx dy = \int_{x=0}^a x \left[\int_{y=0}^{\sqrt{a^2-x^2}} y dy \right] dx$$

$$= \int_{x=0}^a \left(\frac{y^2}{2} \right)_0^{\sqrt{a^2-x^2}} dx = \int_0^a \frac{a^2 - x^2}{2} dx$$

$$= \frac{1}{2} \left[a^2 x - \frac{x^3}{3} \right]_0^a = \frac{1}{2} \left[a^3 - \frac{a^3}{3} \right] = \frac{1}{2} \left[\frac{2a^3}{3} \right] = \frac{a^3}{3}.$$



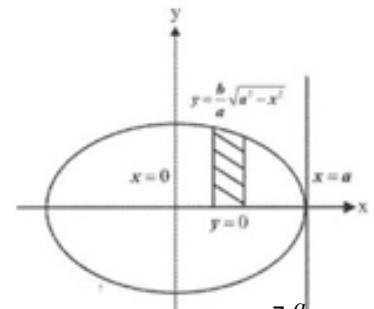
Problem 2. Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution.

Area of the ellipse is

$$= 4 \int_{x=0}^a \int_{y=0}^{\frac{b}{a}\sqrt{a^2-x^2}} dy dx$$

$$= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx = \frac{4b}{a} \left[\frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + \frac{x}{2} \sqrt{a^2 - x^2} \right]_0^a$$



$$\begin{aligned}
 &= \frac{4b}{a} \left[\frac{a^2}{2} \sin^{-1}(1) \right] = \frac{4ba^2}{2a} \frac{\pi}{2} \\
 &= \pi ab
 \end{aligned}$$

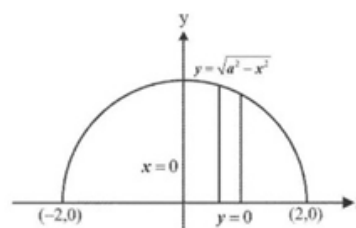
Problem 3. Evaluate $\iint_R dx dy$, where R is the shaded region in the figure

Solution.

Equation of the circle is $x^2 + y^2 = 4$

$\iint_R dx dy = R$, the area of the semi-circle

$$= \frac{1}{2} \pi (2^2) = 2\pi$$



Problem 4. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy(x+y) dx dy$

Solution.

$$\begin{aligned}
 \int_0^1 \int_x^{\sqrt{x}} xy(x+y) dx dy &= \int_{x=0}^1 \left[\int_{y=x}^{\sqrt{x}} (x^2 y + xy^2) dy \right] dx \\
 &= \int_{x=0}^1 \left[\frac{x^2 y^2}{2} + \frac{xy^3}{3} \right]_x^{\sqrt{x}} \\
 &= \int_{x=0}^1 \left[\left(x^2 \left(\frac{x}{2} \right) + \frac{x^2 \sqrt{x}}{3} \right) - \left(\frac{x^4}{2} + \frac{x^4}{3} \right) \right] dx
 \end{aligned}$$

$$\begin{aligned}
&= \int_{x=0}^1 \left(\frac{x^3}{2} + \frac{x^{\frac{5}{2}}}{3} - \frac{5x^4}{6} \right) dx \\
&= \left[\left(\frac{x^4}{8} + \frac{x^{\frac{7}{2}}}{3 \times \frac{7}{2}} - \frac{5x^5}{5 \times 6} \right) \right]_0^1 = \frac{1}{8} + \frac{2}{21} - \frac{1}{6} \\
&= \frac{21 + 16 - 28}{168} = \frac{9}{168} = \frac{3}{56}.
\end{aligned}$$

Problem 5. Evaluate $\int_1^2 \int_0^x \frac{1}{x^2 + y^2} dy dx$

Solution.

$$\begin{aligned}
&\int_{x=0}^2 \left[\int_{y=0}^x \frac{1}{y^2 + x^2} dy \right] dx = \int_{x=1}^2 \left[\frac{1}{x} \tan^{-1} \left(\frac{y}{x} \right) \right]_0^x dx \\
&= \int_{x=1}^2 \frac{1}{x} [\tan^{-1} - \tan^{-1}(0)] dx \\
&= \int_{x=1}^2 \frac{\pi}{4} \frac{1}{x} dx = \frac{\pi}{4} (\log x)_1^2 \\
&= \frac{\pi}{4} [\log 2 - \log 1] = \frac{\pi}{4} \log 2
\end{aligned}$$

Problem 6. Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(\theta + \phi) d\theta d\phi$

Solution.

$$\begin{aligned}
\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(\theta + \phi) d\theta d\phi &= \int_0^{\frac{\pi}{2}} -[\cos(\theta + \phi)]_{\theta=0}^{\frac{\pi}{2}} d\phi \\
&= \int_0^{\frac{\pi}{2}} -\left[\cos\left(\frac{\pi}{2} + \phi\right) - \cos \phi\right] d\phi = \int_0^{\frac{\pi}{2}} (\sin \phi + \cos \phi) d\phi \\
&= (-\cos \phi + \sin \phi)_0^{\frac{\pi}{2}} = \left(-\cos \frac{\pi}{2} + \sin \frac{\pi}{2}\right) \\
&\quad - (-\cos 0 + \sin 0) = 1 + 1 = 2
\end{aligned}$$

Problem 7. Evaluate $\int_0^1 \int_0^z \int_0^{y+z} dz \, dy \, dx$

Solution.

$$\begin{aligned}
\int_{z=0}^1 \int_{y=0}^z \left[\int_{x=0}^{y+z} dx \right] dy \, dz &= \int_{z=0}^1 \int_{y=0}^z [x]_0^{y+z} dy \, dz \\
&= \int_{z=0}^1 \int_{y=0}^z [y + z] dy \, dz = \int_{z=0}^1 \left[\frac{y^2}{2} + zy \right]_0^z dz \\
&= \int_{z=0}^1 \left[\frac{z^2}{2} + z^2 \right]_0^z dz = \int_0^1 \frac{3z^2}{2} dz = \left[\frac{z^3}{2} \right]_0^1 = \frac{1}{2}
\end{aligned}$$

Problem 8. Evaluate $\int_C (x^2 dy + y^2 dx)$ where C is the path $y = x$ from $(0, 0)$ to $(1, 1)$

Solution.

$y = x \Rightarrow dy = dx$ Therefore,

$$\int_c (x^2 dy + y^2 dx) = \int_{x=0}^1 (x^2 dx + x^2 dx) = \int_0^1 2x^2 dx = 2 \left(\frac{x^3}{3} \right)_0^1 = \frac{2}{3}$$

Problem 9. Evaluate $\int_1^b \int_1^a \frac{dx dy}{xy}$

Solution.

$$\begin{aligned} \int_1^b \int_1^a \frac{dx dy}{xy} &= \int_{y=1}^b \frac{1}{y} \left[\int_{x=1}^a \frac{1}{x} dx \right] dy = \int_{y=1}^b \frac{1}{y} (\log x)_1^a dy \\ &= \int_{y=1}^b \frac{1}{y} [\log a - \log 1] dy = \log a \int_1^b \frac{1}{y} dy = \log a (\log y)_1^b \\ &= \log a [\log b - \log 1] = \log a \log b \end{aligned}$$

Problem 10. Write down the double integral to find the area between the circles $r = 2 \sin \theta$ and $r = 4 \sin \theta$

Solution.

$$\text{Required area is} = \int_{\theta=0}^{2\pi} \int_{r=2 \sin \theta}^{4 \sin \theta} r dr d\theta$$

Problem 11. Find the limits of integration in the double integral $\iint_R f(x, y) dx dy$ where R is in the first quadrant bounded by $x = 0, y = 0, x + y = 1$

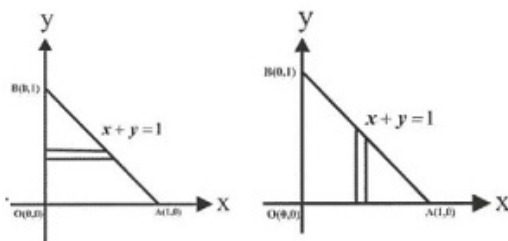
Solution.

Consider a vertical strip as shown in the figure.

Then y varies from 0 to $1 - x$ and x varies from 0 to 1.

(OR)

If we consider a horizontal strip then $x = 0$ to $1 - y$ and $y = 0$ to 1.



Problem 12. Evaluate $\int_0^\pi \int_0^{\sin \theta} r \, dr \, d\theta$

Solution.

$$\begin{aligned} \int_0^\pi \int_0^{\sin \theta} r \, dr \, d\theta &= \int_0^\pi \left(\frac{r^2}{2} \right)_0^{\sin \theta} d\theta = \frac{1}{2} \int_0^\pi \sin^2 \theta d\theta \\ &= \frac{1}{2} \int_0^\pi \frac{(1 - \cos 2\theta)}{2} d\theta \\ &= \frac{1}{4} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^\pi = \frac{1}{4} [\pi] = \frac{\pi}{4} \end{aligned}$$

Problem 13. Express the integral $\iint_R f(x, y) dx dy$ in terms of polar coordinates

Solution.

$$\iint_R f(x, y) dx dy = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta$$

Problem 14. If $x = x(u, v)$ and $y = y(u, v)$, express $dx dy$ in terms of $du dv$ using Jacobian.

Solution.

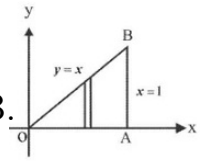
$$dx dy = |J| du dv \text{ where } J = \frac{\partial(x, y)}{\partial(u, v)}$$

Problem 15. Sketch roughly the region of integration for the double integral $\int_0^1 \int_0^x f(x, y) dy dx$

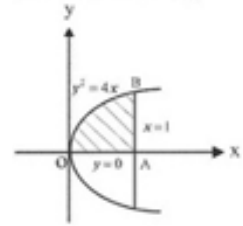
Solution.

$y = 0$ to $y = x$ and $x = 0$ to $x = 1$

$\Rightarrow$ The region of integration is the shaded area OAB.



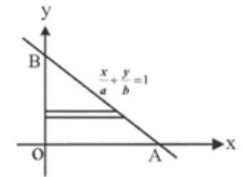
Problem 16. Sketch roughly the region of integration $\iint_R f(x,y) dx dy$, where R is in the first quadrant and bounded by $x = 1, y = 0, y^2 = 4x$

**Solution.**

The region of integration is the shaded area OAB.

Problem 17. Sketch roughly the region of integration

$$\int_0^b \int_0^{\frac{a}{b}(b-y)} f(x,y) dx dy.$$

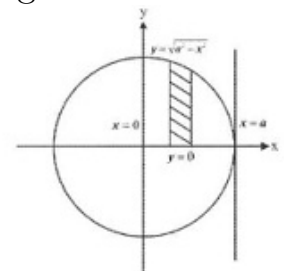
**Solution.**

$x = 0$ to $x = \frac{a}{b}(b-y)$, $\left(\text{i.e., } \frac{x}{a} = 1 - \frac{y}{b} \Rightarrow \frac{x}{a} + \frac{y}{b} = 1 \right)$ $y = 0$ to b

The region of integration is the triangle OAB.

Problem 18. Sketch roughly the region of integration

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} f(x,y) dx dy.$$

**Solution.**

$y = 0$ to $y = \sqrt{a^2 - x^2}$ (i.e., $x^2 + y^2 = a^2$) and $x = 0$ to $x = a$

$\Rightarrow$ the region of integration is the area of the circle in the first quadrant.

Problem 19. Change the order for the double integral

$$\int_0^1 \int_0^x f(x, y) dx dy$$

Solution.

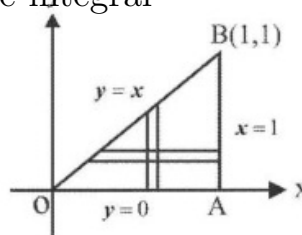
$y = 0$ to $y = x$ and $x = 0$ to $x = 1$

$\Rightarrow$ the area of integration is the triangle OAB with vertical strip considered.

To change the order of integration, consider the horizontal strip.

In this case the limits are $x = y$ to $x = 1$ and $y = 0$ to $y = 1$

$$\text{Thus: } \int_{x=0}^1 \int_{y=0}^x x f(x, y) dy dx = \int_{y=0}^1 \int_{x=y}^1 f(x, y) dx dy$$



Problem 20. Change the order of integration in $I = \int_0^1 \int_{x^2}^{2-x} f(x, y) dx dy$

$$dx dy \int_0^1 \int_0^x f(x, y) dx dy$$

Solution.

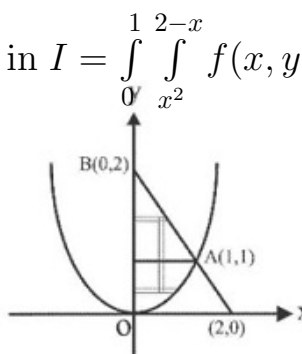
$y = x^2$ to $y = 2 - x$ (or) $x + y = 2$

and $x = 0$ to $x = 1$.

$\Rightarrow$ The region of integration is OAB with vertical strip considered.

Therefore, to change the order of integration consider the horizontal strip.

$$\text{Hence, } I = \int_{y=0}^1 \int_{x=0}^{\sqrt{y}} f(x, y) dx dy + \int_{y=1}^2 \int_{x=0}^{2-y} f(x, y) dx dy$$



Problem 21. Change the order of integration in $\int_0^\infty \int_{\frac{1}{y}} f(x, y) dx dy$

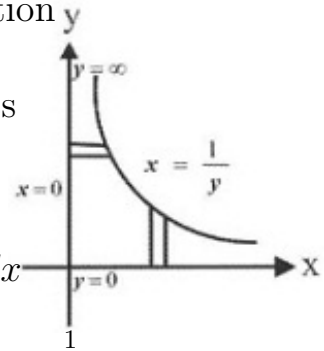
Solution.

$x = 0$ to $x = \frac{1}{y}$ (i.e., $xy = 1$) and $y = 0$ to $y = 0$ to $y = \infty$

$\Rightarrow$ the shaded region is the region of integration with horizontal strip considered.

If we considered vertical strip, then the limits are $y = 0$ to $y = \frac{1}{x}$ and $x = 0$ to $x = \infty$

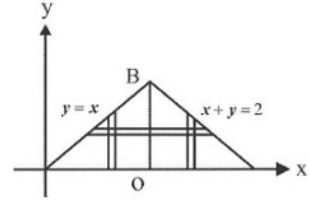
$$\text{Thus, } \int_{y=0}^{\infty} \int_{x=0}^{\frac{1}{y}} f(x, y) dx dy = \int_{x=0}^{\infty} \int_{y=0}^{\frac{1}{x}} f(x, y) dy dx$$



Problem 22. Change the order of integration in $\int_0^1 \int_{\frac{y}{2-y}} f(x, y) dx dy$

Solution.

$x = y$ and $x = 2 - y$ (i.e., $x + y = 2$); $y = 0$ to 1



$\Rightarrow$ the region of integration is the triangle OAB with horizontal strip considered.

To change the order of integration, consider vertical strip. Therefore,

$$\int_{y=0}^1 \int_{x=y}^{2-y} f(x, y) dx dy = \int_{x=0}^1 \int_{y=0}^x f(x, y) dy dx$$

Problem 23. Express $\iiint_V xyz dx dy dz$ in terms of cylindrical polar coordinates

Solution.

$$\begin{aligned} \iiint_V xyz dx dy dz &= \iiint_V r \cos \theta r \sin \theta z r dr d\theta dz \\ &= \iiint_V r^3 \sin \theta \cos \theta dr d\theta dz \end{aligned}$$

Problem 24. Express the transformation from 3 dimensional cartesian to spherical polar coordinates

Solution.

$$x = r \sin \theta \cos \phi, y = r \sin \theta \sin \phi, z = r \cos \theta$$

Problem 25. Express the elementary volume dv in spherical polar coordinates.

Solution.

$$dv = r^2 \sin \theta dr d\theta d\phi$$

UNIT IV

Differential Equations

Problem 1. Show that $(5x^4 + 3x^2y^3 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$ Is an exact equation

$M = 5x^4 + 3x^2y^3 - 2xy^3$ and $2x^3y - 3x^2y^2 - 5y^4$ which implies $\frac{\partial M}{\partial y} = 6x^2y - 6xy^2$ and $\frac{\partial N}{\partial x} = 6x^2y - 6xy^2$. Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence the given equation is exact.

Hence $\int Mdx + Bdy$ (terms in N without x)

i.e., $\int (5x^4 + 3x^2y^3 - 2xy^3)dx + (-5y^4)dy$.

Hence the solution is $x^5 + x^3y^2 - x^2y^3 - y^5 = c$.

Problem 2. Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$.

The given equation can be written as

$$(y \cos x + \sin y + y)dx + (\sin x + x \cos y + x)dy = 0$$

$M = y \cos x + \sin y + y$ and $N = \sin x + x \cos y + x$ which implies

$$\frac{\partial M}{\partial y} = \cos x + \cos y + 1 \text{ and } \frac{\partial N}{\partial x} = \cos x + \cos y + 1.$$

Here $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence the given equation is exact.

Hence $\int Mdx + Ndy$ (terms in without x)

$$\int (y \cos x + \sin y + y)dx + (0)dy = y \sin x + (\sin y + y)x = c$$

Problem 3. Solve $(x^2 - ay)dx = (ax - y^2)dy$

Here $(x^2 - ay)dx - (ax - y^2)dy = 0$. i.e., $(x^2 - ay)dx + (y^2 - ax)dy$.

Therefore $M = x^2 - ay$ and $\frac{\partial M}{\partial y} = -a$. Also $N = y^2 - ax$ and

$$\frac{\partial N}{\partial x} = -a$$

Hence $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. Hence the given equation is exact.

Hence $\int Mdx + Ndy$ (terms in N without x)

$$\int (x^2 - ay)dx + (y^2)dy. \text{ Hence } \frac{x^3}{3} - axy + \frac{y^3}{3} = c.$$

Problem 4. Solve $p^2 - 7p - 12 = 0$. [Nov 2016, May 2016, Jan 2016, May 2014]

The equation is $(p-4)(p-3) = 0$. Hence $p = 4$ or 3 . i.e., $\frac{dy}{dx} = 4$.

Integrating we get

$$y = 4x + c. \text{ Similarly taking } p = 3 \text{ we get } y = 3x + c.$$

$$\text{The solution is } (y - 4x + c)(y - 3x + c) = 0.$$

Problem 5. Write the general solution $y = px + \left(\frac{a}{p}\right)$.

The given equation is of clairauts type $y = px + f(p)$. The general solution is got by writing c for p . Hence the general solution is $y = cx + \left(\frac{a}{c}\right)$.

Problem 6. Solve $p = \sin(y - xp)$. [Jan 2014]

The given equation is $\sin^{-1} p = y - xp$. i.e., $y = px + \sin^{-1} p$. This is clairauts type. The general solution is $y = cx + \sin^{-1} c$.

Problem 7. Solve $p = \log(px - y)$.

The given equation is $e^p = px - y$ or $y = px - e^p$. This is a clairauts type hence the general solution is $y = cx - e^c$.

Problem 8. Solve $y = xdy/dx + adx/dy$. [May 2017, May 2014]

Since $dy/dx = p$. The given equation is $y = px + (a/p)$. The given equation is of clairauts type $y = px + f(p)$. The general solution is got by writing c for p . Hence the general solution is $y = cx + (a/c)$.

Problem 9. Solve $x^2 = 1 - p^2$.

$p^2 = -1 + x^2$. i.e., $p = \pm\sqrt{x^2 - 1}$ which implies $\frac{dy}{dx} = \pm\sqrt{x^2 - 1}$

Hence $dy = \pm\sqrt{x^2 - 1}dx$. Integrating both sides we get

$$y = \pm\frac{x}{2} \pm \log\left(x + \sqrt{x^2 - 1}\right) + c$$

Problem 10. Solve $x = y + a \log p$.

Given $x = y + a \log p$ differentiating the equation with equation with respect to y we get $\frac{dx}{dy} = 1 + \frac{a}{p} \frac{dp}{dy}$. But $\frac{dy}{dx} = p$.

Therefore $\frac{1}{p} = 1 + \frac{a}{p} \frac{dp}{dy}$ which implies $\frac{1}{p} - 1 = \frac{a}{p} \frac{dp}{dy} \Rightarrow \frac{1-p}{p} = \frac{a}{p} \frac{dp}{dy} \Rightarrow 1-p = a \frac{dp}{dy}$

Hence $\frac{dy}{a} = \frac{dp}{1-p}$ Integrating both side of the equation we get

$$\frac{y}{a} - c = -\log(p-1).$$

Hence we get $p = 1 + e^{c-\frac{y}{a}}$. Substituting in the question we get $x = a \log(1 + e^{c-\frac{y}{a}})$ is the solution.

Unit V

Differential Equation Higher Order

Problem 1. Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 13y = 0$. [Similar model in Jan 2015, May 2016, Jan 2016, Nov 2016]

$$(D^2 - 6D + 13)y = 0$$

$$\text{A.E is } m^2 - 6m + 13 = 0$$

$$m = \frac{6 + \sqrt{-16}}{2} = 3 \pm 2i$$

Hence solution is $y = e^{3x}(A \cos 2x + B \sin 2x)$

Problem 2. Solve $(D^2 - 4)y = 4$.

A.E. is $m - 4 = 0$

$$m = 2, -2$$

$$C.F = Ae^{2x} + Be^{-2x}$$

$$P.I = \frac{1}{D^2 - 4}4 = \frac{1}{-4}4.e^{0x} = -1$$

The solution is $y = Ae^{2x} + Be^{-2x} - 1$.

Problem 3. Find the particular of $(D^2 + 5D + 6)y = 11e^{5x}$. [Similar model in May 2014, May 2017]

$$P.I = \frac{11e^{5x}}{D^2 + D + 6}$$

$$\text{Put } D = 5 \text{ hence } P.I = \frac{11e^{5x}}{25 + 5 + 6} = \frac{11e^{5x}}{36}$$

Problem 4. Find the particular of $(D^2 + 4)y = \sin 2x$. [Similar model in Jan 2015]

$$P.I = \frac{\sin 2x}{D^2 + 4}$$

$$\text{Put } D^2 = -2^2, P.I = \frac{\sin 2x}{-4 + 4} = \frac{x \sin 2x}{2D} = \frac{-x}{4} \cos 2x$$

Problem 5. Find the particular of $(D^2 - 2D + 1)y = \cos hx$.

$$P.I = \frac{\cos hx}{D^2 + 2D + 1} = \frac{1}{2} \frac{(e^x + e^{-x})}{D^2 + 2D + 1}$$

$$\begin{aligned} P.I_1 &= \frac{1}{2} \frac{e^x}{D^2 - 2D + 1} = \frac{1}{2} \left(\frac{e^x}{1 - 2 + 1} \right) \\ &= \frac{1}{2} \left(\frac{xe^x}{2D - 2} \right) = \frac{1}{2} \left(\frac{xe^x}{2D - 2} \right) = \frac{1}{2} \left(\frac{xe^x}{2 - 2} \right) = \frac{1}{2} \left(\frac{x^2 e^x}{2} \right) \end{aligned}$$

$$\begin{aligned} P.I_2 &= \frac{1}{2} \frac{e^{-x}}{D^2 + 2D + 1} \\ &= \frac{1}{2} \frac{e^{-x}}{(1 - 2 + 1)} = \frac{1}{2} \frac{xe^{-x}}{2D + 2} = \frac{1}{2} \frac{x^2 e^{-x}}{2} \end{aligned}$$

Problem 6. Find the particular integral of $(D^2 + 2D)y = x^2$. [Similar model in Jan 2014]

$$\begin{aligned} P.I &= \frac{x^2}{D^2 + 2} = \frac{x^2}{2 \left(1 + \frac{D^2}{2} \right)} = \frac{1}{2} \left[1 + \frac{D^2}{2} \right]^{-1} x^2 \\ &= \frac{1}{2} \left[1 - \frac{D^2}{2} + \cdot \right] x^2 = \frac{1}{2} [x^2 - 1] \end{aligned}$$

Problem 7. Find the particular integral of $(D^2 + 1)y = \sin^2 x$.

$$\sin^2 x = \frac{1 - \cos 2x}{2}$$

$$P.I = \frac{\sin^2 x}{D^2 + 1} = \frac{1}{2} \frac{(1 - \cos 2x)}{D^2 + 1} = \frac{1}{2} \left[\frac{1}{D^2 + 1} - \frac{\cos 2x}{D^2 + 1} \right]$$

$$P.I = \frac{1}{1} + \frac{1}{3} \cos 2x$$

Problem 8. Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 3y = \sin 3x \cos 2x$.

$$(D^2 - 4D + 3)y = 0$$

$$\text{A.E is } m^2 - 4m + 3 = 0 \Rightarrow m = 3, m = 1.$$

$$C.F = Ae^{3x} + Be^x$$

$$\sin 3x \cos 2x = \frac{1}{2} [\sin 5x - \sin x]$$

$$P.I. = \frac{1}{2} \frac{\sin 5x}{D^2 + 4D + 3} + \frac{1}{2} \frac{\sin x}{D^2 - 4D + 3}$$

$$P.I._1 = \frac{1}{2} \frac{\sin 5x}{D^2 + 4D + 3}$$

$$P.I._2 = \frac{1}{2} \left(\frac{\sin x}{D^2 + 4D + 3} \right)$$

$$\text{Put } D^2 = -5^2 = -25$$

$$\text{Put } D^2 = -1^2 = -1$$

$$= \frac{1}{2} \frac{\sin 5x}{-22 - 4D}$$

$$= \frac{1}{2} \frac{\sin x}{-4D + 2}$$

$$= \frac{1}{-2} \frac{(22 - 4D) \sin 5x}{22^2 - 16D^2}$$

$$= \frac{1}{2} \frac{(2 + 4D) \sin x}{2^2 - 16D^2}$$

$$P.I. = \frac{1}{884} [10 \cos 5x - 11 \sin 5x] + \frac{1}{20} [\sin x + 2 \cos x]$$

$$y = Ae^{3x} + Be^x + \frac{1}{884} [10 \cos 5x - 11 \sin 5x] + \frac{1}{20} [\sin x + 2 \cos x]$$

Problem 9. Find the particular integral of $(D-3)^2 y = e^{3x} \cos x$ [Similar model in Nov 2016]

$$P.I = \frac{e^{3x} \cos x}{(D-3)^2} = e^{3x} \frac{\cos x}{(D+3-3)^2} = e^{3x} \frac{\cos x}{D^2} = -e^{3x} \cos x$$

Problem 10. Find the particular integral of $(D+2)(D-1)^2 = e^{-2x}$

$$P.I = \frac{1}{(D+2)} \left[\frac{e^{-2x}}{(D-1)^2} \right] = \frac{1}{(D+2)} \left(\frac{e^{-2x}}{9} \right) = \frac{1}{9} \frac{e^{-2x}}{D+2} = \frac{1}{9} \frac{x e^{-2x}}{1}$$

Problem 11. Find the particular integral of $(D^2 + 4D + 4)y = e^{-2x}x$.

$$P.I = \frac{e^{-2x}x}{(D+2)^2} = e^{-2x} \frac{x}{((D-2)+2)^2} = e^{-2x} \frac{1}{D^2}(x) = \frac{e^{-2x}x^3}{6}$$

Problem 12. Write Cauchy's homogenous differential equation.

$$x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n y = X, \text{ where } a_i \text{'s are constants and } X \text{ is a function of } x.$$

Problem 13. Write Lengendre's linear differential equation.

$$(a+bx)^n \frac{d^n y}{dx^n} + A_1 (a+bx)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + A_2 (a+bx)^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + A_n y = f(x), \text{ where } A_i \text{'s are constants.}$$

Problem 14. Write the formula to find the particular integral by method of variation of parameter

$$\text{If } C.F = Ay_1 + By_2$$

$$\text{Then } P.I = Py_1 + Qy_2$$

$$\text{Where } P = \frac{y_2 R}{y_1 y_2 - y_1 y_2} dx; Q = \frac{y_1 R}{y_1 y_2 - y_1 y_2} dx \text{ where } R \text{ is R.H.S.}$$

Problem 15. Solve $x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y = 0$ [Similar model in Jan 2014]

$$(x^2 D^2 + 4x D + 2)y = 0$$

Put $x = e^z, z = \log x, x^2 \frac{d^2 y}{dx^2} = D'(D' - 1)$

$$x \frac{dy}{dx} = D'$$

$$\therefore (D'(D' - 1) + 4D' + 2)z = 0$$

A.E is $m^2 + 3m + 2 = 0 \Rightarrow m = -2, -1.$

$$z = Ae^{-2x} + Be^{-x}$$

$$\Rightarrow y = Ae^{-2(\log x)} + Be^{-(\log x)} \Rightarrow y = \frac{A}{x^2} + \frac{B}{x}$$

Problem 16. Transform the equation $x^2 \frac{d^2 y}{dx^2} + 6x \frac{dy}{dx} + 2y = x \log x$ into linear differential equation with constant coefficients.

$$x = e^z, \log x = z$$

$$x^2 \frac{d^2 y}{dx^2} = D'(D' - 1)$$

$$x \frac{dy}{dx} = D'$$

$$\therefore (D'(D' - 1) + 6D' + 2)z = \Delta e^z$$

Problem 17. Transform the equation $x^2 y'' + xy' = x$ into linear differential equation with constant coefficients

$$(x^2 D^2 + x D)y = x$$

Put $x = e^z, z = \log x, x D =, x^2 D^2 = D'(D' - 1)$

Hence the $(D'(D' - 1) + D')z = x \Rightarrow D'^2 z = x$